

UNIT-II Core Python Assessment

Python Variables and Operators | Control Structures | Python Functions | Strings and String Manipulation

Total Questions	60	Total Marks	100
MCQs	30 x 1 = 30	Fill-ups	20 x 1 = 20
Coding	10 x 5 = 50	Answer Key	Not included in student paper

Instructions

- Answer all questions.
- MCQ and Fill-up questions carry 1 mark each.
- Coding questions carry 5 marks each.
- For coding questions, write only the required function body/code. Follow proper indentation.
- This paper is student-safe and does not contain answers.

Section A - Multiple Choice Questions (30 x 1 = 30 Marks)

Q1. (variables,python-introduction) Who developed Python in 1991?

- A. Dennis Ritchie
- B. Guido van Rossum
- C. James Gosling
- D. Bjarne Stroustrup

Q2. (variables,python-introduction) Python got its name from:

- A. A snake species
- B. Monty Python's Flying Circus
- C. A mathematics theorem
- D. A computer company

Q3. (variables,features) Which feature best describes Python programs according to the notes?

- A. Hard to read
- B. Easily readable and understandable
- C. Only for scientific programming
- D. Platform dependent

Q4. (variables,modes) In interactive mode, the Python prompt generally appears as:

- A. >>>
- B. ###
- C. \$py
- D. :::

Q5. (variables,modes) A Python script file is usually saved with which extension?

- A. .txt
- B. .docx
- C. .py
- D. .java

Q6. (variables,input-output) Which function is used to display output on the screen?

- A. input()
- B. print()
- C. scan()
- D. display()

Q7. (variables,input-output) Which function is used to feed data at runtime?

- A. print()
- B. input()
- C. format()
- D. type()

Q8. (variables,operators) In the expression $5 + 50 * 10$, which operation happens first in Python?

- A. Addition
- B. Multiplication
- C. Subtraction
- D. Assignment

Q9. (control-structures) Which control structure executes statements one after another?

- A. Sequential
- B. Branching
- C. Looping
- D. Recursive

Q10. (control-structures) Branching statements are also called:

- A. Iteration statements
- B. Alternative statements
- C. Sequential statements
- D. Import statements

Q11. (control-if) Simple if is described as a:

- A. Two-way branching statement
- B. One-way branching statement
- C. Looping statement
- D. Function statement

Q12. (control-if-else) An if...else statement contains:

- A. Only true block
- B. Only false block
- C. Both true block and false block
- D. No block

Q13. (control-elif) Which clause should be used at the end of an if...elif...else chain?

- A. if
- B. elif
- C. else
- D. while

Q14. (control-looping) Executing a set of statements repeatedly based on a condition is called:

- A. Branching
- B. Iteration or looping
- C. Compilation
- D. Input

Q15. (control-structures) The three important control structures are:

- A. Input, Output, Process
- B. Sequential, Branching, Looping
- C. Class, Object, Method
- D. String, List, Tuple

Q16. (functions) A function is best described as:

- A. A named block of code for a specific task
- B. A Python file extension
- C. A database table
- D. A string operator

Q17. (functions,advantages) Which is an advantage of functions?

- A. Increase repetition
- B. Avoid repetition and improve reuse
- C. Remove modularity
- D. Prevent parameters

Q18. (functions,types) Which of the following is NOT listed as a type of Python function in the notes?

- A. User-defined function
- B. Built-in function
- C. Lambda function
- D. Database function

Q19. (functions,definition) Which keyword is used to define a function in Python?

- A. let
- B. def
- C. func
- D. define

Q20. (functions,syntax) The function code block comes after which symbol?

- A. comma
- B. semicolon
- C. colon
- D. period

Q21. (functions,return) The statement used to exit from a function is:

- A. break
- B. return
- C. exitonly
- D. stop

Q22. (functions,recursion) A recursion function is a function that:

- A. Never returns
- B. Calls itself
- C. Uses only strings
- D. Has no name

Q23. (strings) A string in Python is used to handle:

- A. Only integers
- B. Array/sequence of characters
- C. Only Boolean values
- D. Only functions

Q24. (strings) Python strings are:

- A. Mutable
- B. Immutable
- C. Always numeric
- D. Only single quoted

Q25. (strings,quotes) A multiline string can be created using:

- A. single quotes only
- B. double quotes only
- C. triple quotes
- D. square brackets

Q26. (strings,indexing) Index value of a string character is also called:

- A. Parameter
- B. Subscript
- C. Argument
- D. Pointer

Q27. (strings,indexing) For the string SCHOOL, the positive index of S is:

- A. 0
- B. 1
- C. -1
- D. 6

Q28. (strings,indexing) For the string SCHOOL, the negative index of the last character L is:

- A. 0
- B. 1
- C. -1
- D. -6

Q29. (strings,operators) The + operator on strings is used for:

- A. Deletion
- B. Concatenation
- C. Division
- D. Comparison only

Q30. (strings,methods) Which method replaces occurrences of a character/string?

- A. delete()
- B. replace()
- C. removechar()
- D. append()

Section B - Fill in the Blanks (20 x 1 = 20 Marks)

Q1. Python is a general purpose _____ language.

Answer: _____

Q2. Python 3.X is used to develop and run _____ code.

Answer: _____

Q3. In _____ mode, Python code is directly typed and executed immediately.

Answer: _____

Q4. A script is a text file that contains Python _____.

Answer: _____

Q5. The _____ function evaluates the expression before printing it.

Answer: _____

Q6. Statements executed one after another are called _____ statements.

Answer: _____

Q7. Branching is also called _____ statement.

Answer: _____

Q8. A simple if statement consists of only the _____ block statements.

Answer: _____

Q9. In an if...elif...else chain, the _____ clause should be used at the end.

Answer: _____

Q10. Repeated execution of statements is called iteration or _____.

Answer: _____

Q11. Functions are named blocks of code used to perform a _____ task.

Answer: _____

Q12. Functions avoid repetition and provide better _____.

Answer: _____

Q13. A lambda function is an anonymous _____ function.

Answer: _____

Q14. A function definition begins with the keyword _____.

Answer: _____

Q15. The function code block should be _____.

Answer: _____

Q16. A string is a sequence of _____ characters.

Answer: _____

Q17. Python strings are _____, meaning they cannot be changed in place.

Answer: _____

Q18. Positive subscript numbers start from 0 to _____.

Answer: _____

Q19. Negative subscript numbers start from -1 to _____.

Answer: _____

Q20. Joining two or more strings is called _____.

Answer: _____

Section C - Simple Coding Questions (10 x 5 = 50 Marks)

Q1. (variables,operators) Write a function `add_two_numbers(a, b)` that returns the sum of a and b.

Starter Code

```
def add_two_numbers(a, b):  
    # write your code here  
    pass
```

Visible Test Cases

Input: `add_two_numbers(5, 10)`
Expected: 15

Input: `add_two_numbers(-3, 8)`
Expected: 5

Write your answer/code:

Q2. (variables,input-output) Write a function `format_sum(x, y)` that returns the string 'The sum of X + Y is Z' using the given values.

Starter Code

```
def format_sum(x, y):  
    # write your code here  
    pass
```

Visible Test Cases

Input: `format_sum(5, 5)`
Expected: The sum of 5 + 5 is 10

Input: `format_sum(2, 3)`
Expected: The sum of 2 + 3 is 5

Write your answer/code:

Q3. (control-if) Write a function `vote_status(age)` that returns 'Eligible to Vote' if age is 18 or above, otherwise returns 'Not Eligible to Vote'.

Starter Code

```
def vote_status(age):  
    # write your code here  
    pass
```

Visible Test Cases

Input: `vote_status(34)`
Expected: Eligible to Vote

Input: `vote_status(14)`
Expected: Not Eligible to Vote

Write your answer/code:

Q4. (control-elif) Write a function `grade_label(mark)` that returns 'A' for `mark >= 80`, 'B' for `mark >= 60`, 'C' for `mark >= 40`, otherwise 'Fail'.

Starter Code

```
def grade_label(mark):  
    # write your code here  
    pass
```

Visible Test Cases

Input: `grade_label(85)`
Expected: A

Input: `grade_label(63)`
Expected: B

Input: `grade_label(35)`
Expected: Fail

Write your answer/code:

Q5. (functions) Write a function `square(x)` that returns the square of x.

Starter Code

```
def square(x):  
    # write your code here  
    pass
```

Visible Test Cases

Input: `square(5)`
Expected: 25

Input: `square(12)`
Expected: 144

Write your answer/code:

Q6. (functions,return) Write a function `greet()` that returns 'Welcome to Python'.

Starter Code

```
def greet():  
    # write your code here  
    pass
```

Visible Test Cases

Input: `greet()`
Expected: Welcome to Python

Write your answer/code:

Q7. (functions,recursion) Write a recursive function factorial(n) that returns factorial of n. Assume n is a non-negative integer.

Starter Code

```
def factorial(n):  
    # write your code here  
    pass
```

Visible Test Cases

Input: factorial(0)

Expected: 1

Input: factorial(5)

Expected: 120

Write your answer/code:

Q8. (strings,indexing) Write a function first_last(text) that returns the first and last characters joined together.

Starter Code

```
def first_last(text):  
    # write your code here  
    pass
```

Visible Test Cases

Input: first_last('SCHOOL')

Expected: SL

Input: first_last('Python')

Expected: Pn

Write your answer/code:

Q9. (strings,methods) Write a function replace_a(text) that replaces every uppercase 'A' with '*' and returns the new string.

Starter Code

```
def replace_a(text):  
    # write your code here  
    pass
```

Visible Test Cases

Input: replace_a('AAAbc')

Expected: ***bc

Input: replace_a('BANANA')

Expected: B*N*N*

Write your answer/code:

Q10. (strings,operators) Write a function `join_words(a, b)` that returns two strings concatenated with one space between them.

Starter Code

```
def join_words(a, b):  
    # write your code here  
    pass
```

Visible Test Cases

Input: `join_words('welcome', 'Python')`

Expected: `welcome Python`

Input: `join_words('Hello', 'World')`

Expected: `Hello World`

Write your answer/code:
